

Calculus and Linear Algebra

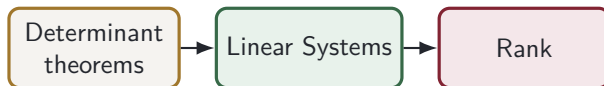
Determinant theorems and Linear Systems

Lecture 6

Daniele Angelini
Sapienza University of Rome

28 September 2026

Lecture outline



Aim

The study of a matrix is useful to solve linear systems. We will introduce the rank to understand the linear (in)dependence contained in a linear system.

A 4×4 determinant

Laplace expansion is recursive: an $n \times n$ determinant is reduced to determinants of order $n - 1$.

For example,

$$A = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 0 & 0 \\ 4 & 1 & 2 & 0 \\ 0 & 0 & 5 & 1 \end{pmatrix}.$$

The second row contains three zeros, so expanding along it reduces the problem immediately to one 3×3 determinant.

Exercise: a sparse 4×4 matrix

Exercise

Compute

$$\det \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 0 & 0 \\ 4 & 1 & 2 & 0 \\ 0 & 0 & 5 & 1 \end{pmatrix}$$

by Laplace expansion.

Indicate clearly the row or column used at each step.

Exercise: solution

Solution Expanding along row 2,

$$\det A = 3 \det \begin{pmatrix} 1 & 0 & 2 \\ 4 & 2 & 0 \\ 0 & 5 & 1 \end{pmatrix}.$$

Expanding the 3×3 determinant along the first row,

$$\det A = 3 \left[1 \det \begin{pmatrix} 2 & 0 \\ 5 & 1 \end{pmatrix} + 2 \det \begin{pmatrix} 4 & 2 \\ 0 & 5 \end{pmatrix} \right] = 3(2 + 40) = \boxed{126}.$$

Triangular matrices

Theorem

If A is upper triangular or lower triangular, then

$$\det A = \prod_{i=1}^n a_{ii}.$$

Example:

$$A = \begin{pmatrix} 2 & 1 & 4 \\ 0 & -3 & 5 \\ 0 & 0 & 6 \end{pmatrix} \implies \det A = 2(-3)6 = -36.$$

Immediate cases

For every n ,

$$\det I_n = 1.$$

For the zero matrix,

$$\det 0 = 0.$$

More generally, any triangular matrix with at least one zero on the main diagonal has determinant zero. **Exercise** Compute mentally:

$$\det \begin{pmatrix} 4 & 2 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & -2 \end{pmatrix}, \quad \det \begin{pmatrix} 1 & 0 & 0 \\ 5 & 0 & 0 \\ 2 & 4 & 3 \end{pmatrix}.$$

Effect of exchanging two rows

Theorem

If B is obtained from A by interchanging two rows, then

$$\det B = -\det A.$$

The same statement holds for columns.

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \det A = -2,$$

while

$$B = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}, \quad \det B = 2.$$

Multiplying one row by a scalar

Theorem

If one row of A is multiplied by c , then the determinant is multiplied by c .

$$R_i \mapsto cR_i \implies \det B = c \det A.$$

Consequently, multiplying the whole $n \times n$ matrix by c gives

$$\det(cA) = c^n \det A.$$

Adding a multiple of another row

Theorem

The operation

$$R_i \mapsto R_i + cR_j, \quad i \neq j,$$

does not change the determinant:

$$\det B = \det A.$$

This property will be important when row operations are introduced for linear systems.

Determinant of a product

Theorem

For square matrices A and B of the same order,

$$\det(AB) = \det A \det B.$$

In general,

$$\det(A + B) \neq \det A + \det B.$$

The determinant is multiplicative, not additive.

Transpose and determinant

Theorem

For every square matrix A ,

$$\det(A^T) = \det A.$$

Thus every theorem stated in terms of rows has an analogous formulation in terms of columns. In particular,

$$\text{dependent rows} \iff \text{dependent columns}$$

for a square matrix with determinant zero.

Determinant properties: summary

Operation / relation	Effect on determinant
swap two rows	changes sign
$R_i \mapsto cR_i$	multiplied by c
$R_i \mapsto R_i + cR_j$	unchanged
transpose	unchanged
product AB	$\det A \det B$
triangular matrix	product of diagonal entries

Exercise: use properties, not expansion

Exercise

Suppose A is a 3×3 matrix with

$$\det A = -4.$$

Find:

1. $\det(A^\top)$;
2. $\det(2A)$;
3. $\det(A^2)$;
4. the determinant after swapping two rows of A ;
5. the determinant after replacing R_3 by $R_3 + 5R_1$.

Exercise: solution

Solution Since A is 3×3 and $\det A = -4$,

$$\det(A^T) = -4,$$

$$\det(2A) = 2^3(-4) = -32,$$

$$\det(A^2) = (-4)^2 = 16.$$

After one row swap, the determinant is 4.

After $R_3 \mapsto R_3 + 5R_1$, the determinant remains -4 .

Determinant and linear dependence

Theorem

Let A be an $n \times n$ matrix. Then

$$\det A = 0 \iff \text{the columns of } A \text{ are linearly dependent.}$$

Equivalently,

$$\det A = 0 \iff \text{the rows of } A \text{ are linearly dependent.}$$

A structural zero determinant

Consider

$$A = \begin{pmatrix} 1 & 2 & 4 \\ 3 & 0 & 6 \\ 2 & 4 & 8 \end{pmatrix}.$$

The third row is twice the first row:

$$R_3 = 2R_1.$$

Hence the rows are linearly dependent, and therefore

$$\boxed{\det A = 0}$$

without any Laplace expansion.

Exercise: detect zero determinants

Exercise

Determine which matrices have determinant zero without computing all determinants explicitly.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 3 & 1 \\ 4 & 1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 6 & 1 \\ -1 & -2 & 4 \end{pmatrix}.$$

State the reason in each case.

Exercise: solution

Solution For A ,

$$R_2 = 2R_1,$$

so $\det A = 0$.

For C , the first two columns are dependent:

$$C_2 = 2C_1,$$

so $\det C = 0$.

Matrix B has no immediate dependence relation; direct computation gives

$$\det B = -25 \neq 0.$$

From equations to linear systems

A **linear system** is a collection of linear equations that must hold simultaneously.

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1, \\ a_{21}x_1 + \cdots + a_{2n}x_n = b_2, \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m. \end{cases}$$

Two numbers matter immediately

m = number of equations,

n = number of unknowns.

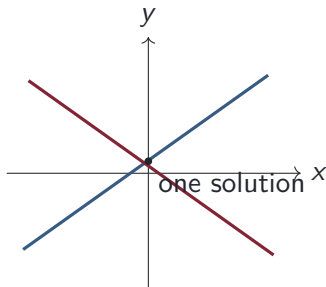
A system may therefore be square ($m = n$), overdetermined ($m > n$), or underdetermined ($m < n$).

The 2×2 case

Consider

$$\begin{cases} a_{11}x + a_{12}y = b_1, \\ a_{21}x + a_{22}y = b_2. \end{cases}$$

Each equation represents a line in the plane.



The three possibilities are:

- ▶ one intersection $\Rightarrow$ one solution;
- ▶ parallel distinct lines $\Rightarrow$ no solution;
- ▶ the same line $\Rightarrow$ infinitely many solutions.

A 3×2 system: more equations than unknowns

$$\begin{cases} x + y = 2, \\ 2x - y = 1, \\ 3x = 3. \end{cases}$$

Here $m = 3$ and $n = 2$.

Geometrically, we have **three lines** in $\mathbb{R}^2$.
A common solution must belong to all three lines.

Important

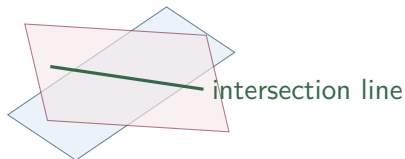
Having more equations than unknowns does *not* automatically mean that the system has no solution.

The real question is whether the equations are mutually compatible.

A 2×3 system: more unknowns than equations

$$\begin{cases} x + y + z = 2, \\ 2x - y + z = 1. \end{cases}$$

Here $m = 2$ and $n = 3$. Each equation represents a plane in $\mathbb{R}^3$.



Generically, two non-parallel planes intersect along a line: infinitely many solutions.

The 3×3 case

A system with three equations and three unknowns has the form

$$\begin{cases} a_{11}x + a_{12}y + a_{13}z = b_1, \\ a_{21}x + a_{22}y + a_{23}z = b_2, \\ a_{31}x + a_{32}y + a_{33}z = b_3. \end{cases}$$

Each equation is a plane in $\mathbb{R}^3$. Possible common intersections include:

- ▶ one point;
- ▶ a line;
- ▶ a plane;
- ▶ no common point.

The algebraic language of matrices will let us classify all these cases systematically.

The generic $m \times n$ linear system

The general system

$$\sum_{j=1}^n a_{ij}x_j = b_i, \quad i = 1, \dots, m,$$

contains m equations and n unknowns. It is completely described by three objects:

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}.$$

Matrix form: $Ax = b$

The whole system can be written compactly as

$$Ax = b.$$

Indeed,

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}.$$

A is called *Coefficients matrix*. b is the vector of constant terms.

Dimensions

If $A \in \mathbb{R}^{m \times n}$, then $x \in \mathbb{R}^n$ and $b \in \mathbb{R}^m$.

Example: from equations to matrices

Exercise Write the system

$$\begin{cases} 2x - y + 3z = 7, \\ x + 4y - z = 0, \\ -3x + 2z = 5 \end{cases}$$

in matrix form. **Solution**

$$\underbrace{\begin{pmatrix} 2 & -1 & 3 \\ 1 & 4 & -1 \\ -3 & 0 & 2 \end{pmatrix}}_A \underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_x = \underbrace{\begin{pmatrix} 7 \\ 0 \\ 5 \end{pmatrix}}_b.$$

Notice that a missing variable corresponds to a zero coefficient.

The augmented matrix

For solving a system, it is useful to place the vector b next to the coefficient matrix A :

$$\boxed{[A \mid b]}.$$

For the previous example,

$$[A \mid b] = \left(\begin{array}{ccc|c} 2 & -1 & 3 & 7 \\ 1 & 4 & -1 & 0 \\ -3 & 0 & 2 & 5 \end{array} \right).$$

The vertical bar is only a visual separator: it distinguishes coefficients from constant terms.

What are our objectives?

Given a linear system $Ax = b$, we want to answer three questions.

1. **Existence:** does at least one solution exist?
2. **Uniqueness:** if a solution exists, is it unique?
3. **Description:** if there are infinitely many solutions, how many free parameters are needed?

Key idea

The answer is controlled not only by the number of equations and unknowns, but by the amount of **independent information** contained in the equations.

That amount of independent information is measured by the **rank**.

Why counting equations is not enough

Compare

$$\begin{cases} x + y = 2, \\ 2x + 2y = 4 \end{cases} \quad \text{and} \quad \begin{cases} x + y = 2, \\ 2x + 2y = 5. \end{cases}$$

Both systems have two equations and two unknowns. But in the first system, the second equation repeats the information of the first; in the second system, it contradicts it.

Lesson

We need a way to detect how many rows or columns carry genuinely independent information.

Rank of a matrix: intuition

The **rank** of a matrix is the number of independent directions encoded by its rows or columns. For example,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{pmatrix}.$$

The second row is twice the first, so it adds no new independent information. In this case the matrix has rank 2:

$$\boxed{\text{rank}(A) = 2.}$$

Formal definition using minors

Definition. The rank of a matrix A is the largest order of a non-zero minor of A .

$$\text{rank}(A) = r$$

means that:

- ▶ at least one $r \times r$ minor has non-zero determinant;
- ▶ every minor of order $r + 1$ is zero.

Therefore, for $A \in \mathbb{R}^{m \times n}$,

$$0 \leq \text{rank}(A) \leq \min\{m, n\}.$$

Rank of a 2×2 matrix

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Then:

$$\text{rank}(A) = \begin{cases} 2, & \det A \neq 0, \\ 1, & \det A = 0 \text{ and } A \neq 0, \\ 0, & A = 0. \end{cases}$$

Thus, for a square 2×2 matrix, the determinant immediately tells us whether the rank is maximal.

Rank of a 3×3 matrix

For $A \in \mathbb{R}^{3 \times 3}$:

- ▶ if $\det A \neq 0$, then $\text{rank}(A) = 3$;
- ▶ if $\det A = 0$, inspect the 2×2 minors;
- ▶ if at least one 2×2 minor is non-zero, then $\text{rank}(A) = 2$;
- ▶ if all 2×2 minors vanish but $A \neq 0$, then $\text{rank}(A) = 1$;
- ▶ if $A = 0$, then $\text{rank}(A) = 0$.

Strategy

Start from the largest possible minors and move downward only when necessary.

Exercise: compute a rank with minors

Exercise Determine the rank of

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 0 \end{pmatrix}.$$

Since $R_2 = 2R_1$, we know immediately that

$$\det A = 0,$$

so $\text{rank}(A) < 3$. Now find one non-zero 2×2 minor.

Exercise: solution

Solution Consider the minor obtained from rows 1, 3 and columns 1, 2:

$$\det \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} = 1 - 2 = -1 \neq 0.$$

Therefore,

$$\text{rank}(A) \geq 2.$$

But $\det A = 0$, hence $\text{rank}(A) < 3$. Thus

$$\boxed{\text{rank}(A) = 2.}$$

Row rank and column rank

A fundamental theorem states that

$$\text{row rank}(A) = \text{column rank}(A).$$

Therefore we simply speak of **the rank** of A . Consequences:

- ▶ rank counts the maximum number of linearly independent rows;
- ▶ equivalently, it counts the maximum number of linearly independent columns.

This connects the definition by minors with linear dependence.

Rank and elementary row operations

Elementary row operations do not change the rank:

$$R_i \leftrightarrow R_j, \quad R_i \mapsto cR_i \ (c \neq 0), \quad R_i \mapsto R_i + cR_j.$$

Hence

$$\boxed{\text{rank}(A) = \text{rank}(EA)}$$

for every elementary matrix E . This gives a second practical method for computing rank: reduce the matrix to echelon form and count the non-zero rows.

Rank by row reduction: example

Consider

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 0 \end{pmatrix}.$$

Apply

$$R_2 \mapsto R_2 - 2R_1, \quad R_3 \mapsto R_3 - R_1.$$

Then

$$A \sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -1 & -3 \end{pmatrix}.$$

There are two non-zero rows, so

$$\boxed{\text{rank}(A) = 2.}$$

Sylvester's rank inequality

Theorem (Sylvester)

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. Then

$$\text{rank}(A) + \text{rank}(B) - n \leq \text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}.$$

The lower bound is known as **Sylvester's rank inequality**. In particular, multiplying matrices cannot create more independent directions than either factor already contains.

Full rank

A matrix $A \in \mathbb{R}^{m \times n}$ has **full rank** when

$$\text{rank}(A) = \min\{m, n\}.$$

Therefore:

- ▶ if $m \geq n$, full rank means $\text{rank}(A) = n$;
- ▶ if $m \leq n$, full rank means $\text{rank}(A) = m$.

For a square matrix $A \in \mathbb{R}^{n \times n}$,

$A \text{ full rank} \iff \text{rank}(A) = n \iff \det A \neq 0.$

This is the bridge between determinants and linear systems.

Rank and linear systems: the bridge

For the system

$$Ax = b,$$

we compare the rank of the coefficient matrix A with the rank of the augmented matrix $[A \mid b]$.



This comparison will tell us whether the system is compatible and whether the solution is unique or depends on free parameters.

Preparation for Lecture 7

Next time we will use rank to classify and solve linear systems.

- ▶ Gaussian elimination and echelon form;
- ▶ compatible versus incompatible systems;
- ▶ the Rouché–Capelli theorem;
- ▶ unique solutions and infinitely many solutions;
- ▶ free variables and parametric form of the solution set.

Main question

Given $Ax = b$, how can the ranks of A and $[A \mid b]$ determine the complete structure of the solution set?



“If man did not know mathematics, he would not rise a single palm above the ground.”

René Descartes

